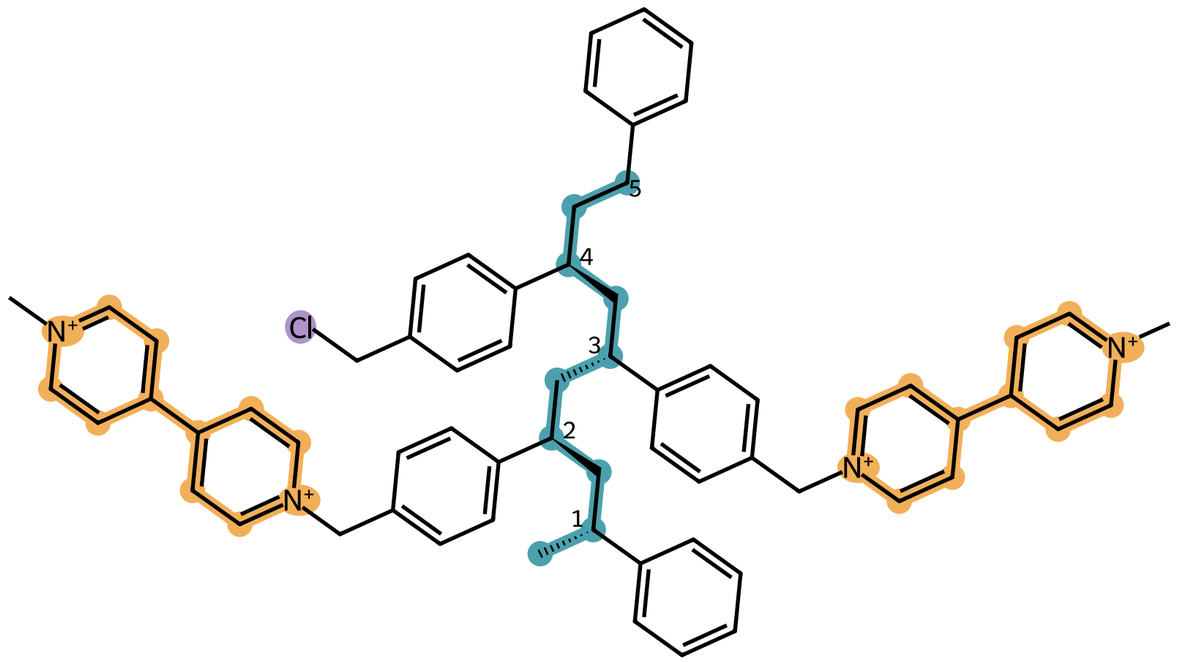
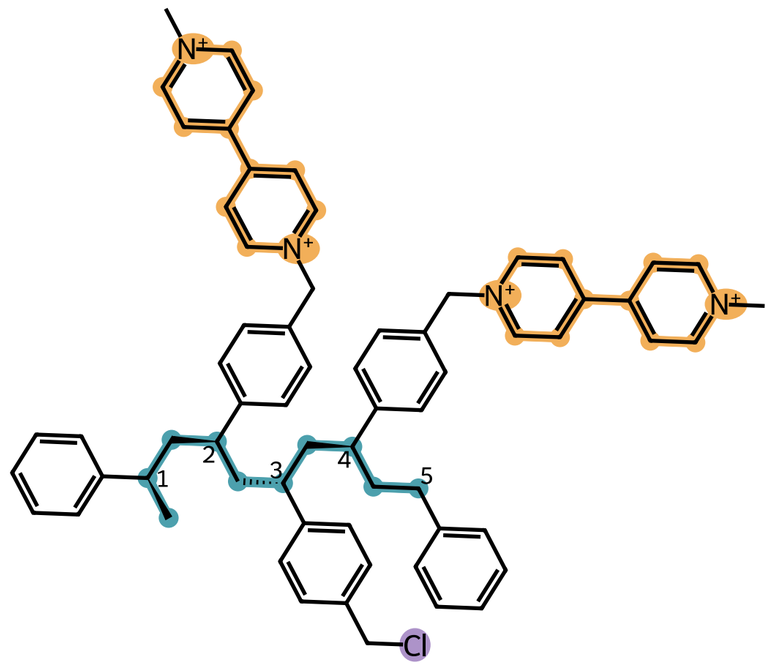
Exact viologen on an open five-repeat polymer model

Same handle positions (2, 3, 4); only occupied sites change. Two grafts give bare charge +4.

Adjacent grafts: sites 2 + 3



One spacer: sites 2 + 4



Teal: open carbon backbone · Orange: viologen cores · Purple: residual benzyl chloride

Numbers label repeat sites. Terminal repeats 1 and 5 are unfunctionalized styrene padding.

Connectivity drawings; prepared XYZ structures are audited separately. No interaction energy is shown.

Counterions are omitted from coordinates. PF_6^- appears only in a separate dry-salt mass convention.